

Functional Annotation Report: SelA (L-seryl-tRNA(Sec) selenium transferase) in *Pseudomonas putida* KT2440

Gene: `selA` (Ordered locus **PP_0493**) **UniProt:** Q88QJ8 · **Organism:** *Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440) **EC** 2.9.1.1 · **Family:** SelA family; fold-type-I pyridoxal-5'-phosphate (PLP)-dependent transferase superfamily **Synonyms:** Selenocysteine synthase; Sec synthase; Selenocysteinyl-tRNA(Sec) synthase **Date:** 2026-07-27

0. Identity verification (mandatory)

The requested target is **SelA of *P. putida* KT2440 (Q88QJ8, PP_0493)**. The UniProt descriptors — gene symbol *selA*, EC 2.9.1.1, "L-seryl-tRNA(Sec) selenium transferase / selenocysteine synthase", membership in the **SelA family**, and the diagnostic domain set (PyrdxLP-dependent transferase IPR015421/IPR015424; **SelA_trans** IPR004534; **SelA_trans_N_dom** IPR025862; SelA-like IPR018319) — are all mutually consistent and unambiguously identify a **bacterial selenocysteine synthase**.

There is no significant gene-symbol ambiguity: in bacterial genetics *selA* has a single, stable meaning (the selenocysteine synthase of the *sel* operon), distinct from the eukaryotic/archaeal counterpart **SepSecS** (which uses a different substrate). Direct experimental literature exists for orthologs in *Escherichia coli*, *Aquifex aeolicus*, and *Moorella thermoacetica*; the *P. putida* protein is annotated by homology (HAMAP rule MF_00423). Because SelA is a highly conserved housekeeping enzyme with essentially one biochemical function across bacteria, the mechanistic and structural conclusions from these orthologs transfer directly to the *P. putida* protein. All function-level claims below are therefore made with high confidence; strain-specific *P. putida* biochemical studies of PP_0493 itself are not available and this is noted as a limitation.

1. Summary (answer to the research question)

SelA (PP_0493) is a pyridoxal-5'-phosphate (PLP)-dependent selenocysteine synthase (EC 2.9.1.1) that catalyzes the terminal, committed step of selenocysteine (Sec, the 21st amino acid) biosynthesis in the bacterial cytoplasm. It converts **L-seryl-tRNA(Sec)** into **L-selenocysteinyl-tRNA(Sec)**, replacing the serine hydroxyl with a selenol group using **selenophosphate** as the activated selenium donor. Its substrate is not a free amino acid but an **aminoacylated tRNA(Sec)**, and its exquisite specificity for tRNA(Sec) (over the near-identical Ser-tRNA(Ser)) is enforced by a large **homodecameric (≈ 500 kDa) quaternary structure** whose N-terminal domains read the tRNA(Sec)-specific D-arm. The charged Sec-tRNA(Sec) it produces is handed to the dedicated elongation factor **SelB**, which recodes in-frame UGA codons (via SECIS elements) to insert Sec co-translationally into bacterial selenoproteins.

2. Primary molecular function — the catalyzed reaction

2.1 Reaction and substrate specificity

SelA catalyzes:

L-seryl-tRNA(Sec) + selenophosphate $\rightarrow$ L-selenocysteinyl-tRNA(Sec) + phosphate
(PLP-dependent).

The pathway context: tRNA(Sec) (the product of *selC*) is first charged with **serine** by the canonical seryl-tRNA synthetase, producing Ser-tRNA(Sec); SelA then converts this to Sec-tRNA(Sec) [PMID 1839607; 23266652]. Critically, **SelA acts on the tRNA-bound amino acid, not free serine**, and it specifically recognizes tRNA(Sec): "Selenocysteine synthase catalyses the synthesis of selenocysteyl-tRNA(Sec) from seryl-tRNA(Sec) in a pyridoxal phosphate-dependent reaction mechanism. The enzyme specifically recognizes the tRNA(Sec) molecule" [PMID 1839607]. The *E. coli* enzyme is formally defined as "Selenocysteine Synthase (SELA, E.C. 2.9.1.1) ... responsible for the conversion of seryl-tRNA(sec) into selenocysteyl-tRNA(sec)" [PMID 23266652].

2.2 Cofactor and chemical mechanism

SelA is a **PLP (vitamin B6)-dependent enzyme** of the **fold-type-I PLP superfamily** [PMID 24456689]. Mechanistically, PLP forms a Schiff-base (aldimine) with the α -amino group of the tRNA-bound seryl residue; β -elimination of the serine hydroxyl generates a PLP-stabilized **dehydroalanyl (aminoacrylyl)-tRNA** intermediate, to which the selenol nucleophile derived from **selenophosphate** is added, yielding selenocysteinyl-tRNA(Sec). Structural work on *Aquifex aeolicus* SelA shows that "SelA catalyzes pyridoxal 5'-phosphate-dependent Sec formation involving **Arg residues** nonhomologous to those in SepSecS" [PMID 23559248] — i.e., bacterial SelA uses a distinct constellation of catalytic residues from the archaeal/eukaryotic enzyme, evidence that the two Sec-synthesis systems arose by **independent (convergent) evolution** despite both being PLP enzymes.

2.3 Selenium donor

The activated selenium substrate is **selenophosphate**, produced by **selenophosphate synthetase (SelD)** from selenide and ATP. This is conserved across all three domains of life: the eukaryotic/archaeal counterpart likewise "required selenophosphate and O-phosphoseryl-tRNA([Ser]Sec) as substrates to generate selenocysteyl-tRNA([Ser]Sec)" [PMID 17194211]. The key bacterial-vs-eukaryotic distinction is the serine intermediate: **bacterial SelA acts directly on Ser-tRNA(Sec)**, whereas eukaryotic/archaeal SepSecS first requires phosphorylation of the seryl moiety to O-phosphoseryl-tRNA(Sec) [PMID 18093968; 17194211].

3. Structure–function: how specificity is achieved

- **Quaternary structure.** SelA "has an exceptional homodecameric quaternary structure with a molecular mass of about 500kDa" [PMID 24456689]. The *E. coli* enzyme is a homodecamer of ~500 kDa ($pI \approx 6.03$), assembling by stepwise addition of oligomeric intermediates rather than a direct monomer→decamer jump [PMID 23266652].
- **A pentamer of dimers.** The ring "is composed of pentamerized SelA dimers," and the **catalytic site lies close to the dimer–dimer interface**. Engineered "depentamerized" (dimeric) variants "formed a distorted and inactivated catalytic site," proving that higher-order assembly is required to build a productive active site [PMID 24456689].
- **tRNA recognition and substrate discrimination.** The decamer "binds 10 tRNA(Sec) molecules, each interacting with four SelA subunits," and "The SelA **N-terminal domain**

binds the tRNA(Sec)-specific D-arm structure, thereby discriminating Ser-tRNA(Sec) from Ser-tRNA(Ser)" [PMID 23559248]. This corresponds to the InterPro **SelA_trans_N_dom (IPR025862)** annotated in Q88QJ8. Binding stoichiometry is ~1 tRNA(Sec) per SelA monomer with D5 point-group symmetry [PMID 23428847].

- **Decamerization is functionally essential**, confirmed by in vivo and in vitro assays [PMID 23559248].

Together these establish that the large ring is the functional unit; oligomerization simultaneously creates the composite catalytic sites and the extended surface that reads tRNA(Sec) identity elements, guaranteeing that only the correct tRNA is selenylated.

3b. Organism-specific confirmation in *P. putida* KT2440 (bioinformatic evidence)

Direct inspection of the UniProt/genomic record for the target protein and its genomic neighborhood corroborates every functional claim at the level of *P. putida* KT2440 itself (not just orthologs):

- **Q88QJ8 (PP_0493)** is a **475-residue** protein carrying the diagnostic **PLP Schiff-base lysine at position 297** (annotated "N6-(pyridoxal phosphate)lysine") — the catalytic residue expected for a fold-type-I PLP enzyme.
- Annotated **catalytic activity (Rhea:22728)**: *L*-seryl-tRNA(Sec) + selenophosphate + H^+ = *L*-selenocysteinyl-tRNA(Sec) + phosphate; **cofactor**: pyridoxal 5'-phosphate; **pathway**: "selenocysteinyl-tRNA(Sec) from L-seryl-tRNA(Sec) (bacterial route), step 1/1"; **subcellular location**: Cytoplasm.
- **Genomic clustering — a complete, co-localized Sec machinery**. *selA* (PP_0493) is **immediately adjacent to selB (PP_0494)**, reproducing the *E. coli* *selAB* operon arrangement. The upstream neighborhood encodes a **formate-dehydrogenase operon: PP_0489 (fdoG, 1022 aa)** — the catalytic major subunit that **contains a genetically encoded selenocysteine at position 197** (UniProt "Non-standard residue: Selenocysteine") — together with **PP_0490** (formate dehydrogenase iron-sulfur subunit) and **PP_0492** (FdH_E maturation homolog). The selenium-donor enzyme **SelD / selenophosphate synthetase (PP_0823)**, which "synthesizes selenophosphate from selenide and ATP," resides elsewhere in the genome.

Interpretation: This is direct, organism-specific evidence for SelA's role. *P. putida* KT2440 encodes a bona-fide selenoprotein (FdoG, Sec-197). The in-frame UGA that specifies that Sec can only be translated if SelA has generated **Sec-tRNA(Sec)**. Thus SelA (PP_0493) is functionally required to mature the cell's **selenocysteine-dependent formate dehydrogenase**, and the physical clustering of *selAB* with the *fdo* formate-dehydrogenase genes ties the enzyme to a concrete downstream client rather than a merely inferred one.

3c. Sequence/evolutionary evidence — an intact, canonical active site

A global pairwise alignment shows ***P. putida* SelA (Q88QJ8, 475 aa) is 65.2% identical (283/434 aligned positions) to *E. coli* SelA (P0A821)** — the enzyme experimentally validated as a PLP-dependent, ~500 kDa homodecameric selenocysteine synthase. Critically, the **catalytic PLP-Schiff-base lysine is conserved within an invariant fold-type-I motif**: *P. putida* `VTF**SGDK297LLGGPQ**` vs *E. coli* `VSF**SGDKLLGGPQ**` (only a conservative T/S difference outside the core motif). This is fully consistent with UniProt's independent annotation of **Lys297** as the N6-(pyridoxal phosphate)lysine.

Interpretation: The high whole-length identity to a functionally proven ortholog, plus strict conservation of the catalytic lysine and its signature motif, is strong evolutionary evidence that the *P. putida* enzyme has an intact, canonical selenocysteine-synthase active site. It is a genuine SelA ortholog — not a divergent or pseudo-enzymatic paralog — which justifies transferring the *E. coli/A. aeolicus* mechanistic and structural conclusions to this protein.

4. Localization

SelA is a **soluble cytoplasmic (cytosolic) enzyme**. It has no signal peptide, no transmembrane segments, and acts on a cytoplasmic substrate (an aminoacyl-tRNA) in a process tightly coupled to ribosomal translation. Recombinant *E. coli* SelA is purified as a soluble ~500 kDa protein by standard chromatography [PMID 23266652]. Its product is consumed in the cytoplasm by SelB during translation [PMID 1839607; 12486013]. This is consistent with the UniProt/HAMAP annotation of a cytoplasmic location.

5. Pathway context — the bacterial selenocysteine incorporation machinery

SelA is one of four dedicated components (plus seryl-tRNA synthetase) that recode UGA to insert Sec:

Gene	Product	Role relative to SelA
<i>selC</i>	tRNA(Sec)	Scaffold substrate; charged with Ser, then converted by SelA
<i>selD</i>	Selenophosphate synthetase	Makes the selenium donor (selenophosphate) used by SelA
<i>selA</i>	Selenocysteine synthase (this protein)	Converts Ser-tRNA(Sec) → Sec-tRNA(Sec)
<i>selB</i>	Sec-specific elongation factor (EF-Tu-like)	Accepts SelA's product; decodes UGA at SECIS elements

- SelA's product is channeled to SelB: "SELB is an EF-Tu-like protein which specifically complexes selenocysteyl-tRNA(Sec)... Interaction with the selenol group of the side chain ... is a prerequisite for the formation of a stable SELB-tRNA complex" [PMID 1839607]. This selenol-dependent recognition is the checkpoint ensuring only fully synthesized Sec-tRNA(Sec) (SelA's output) is used.
- Downstream, the **SECIS element** immediately 3' of the UGA "tether[s] a complex of the selenocysteine-specific elongation factor SelB, GTP and selenocysteyl-tRNA(Sec) to the site of UGA decoding" [PMID 12486013].
- **Operon organization and regulation.** In *E. coli*, "The **selAB operon** codes for the proteins selenocysteine synthase and SELB" [PMID 1839607], and a SECIS-like element in the *selAB* leader couples expression of the pathway to selenium availability [PMID 12486013]. This *selAB* arrangement is **conserved in *P. putida***, where *selA* (PP_0493) and *selB* (PP_0494) are immediately adjacent (§3b); *selD* (PP_0823) lies elsewhere in the genome.
- **Biological end-products.** The Sec-tRNA(Sec) generated by SelA is used to synthesize bacterial **selenoproteins**, canonically the anaerobic/formate-linked oxidoreductases (e.g., **formate dehydrogenases**) and related redox enzymes, where Sec provides a highly reactive catalytic selenol. In *P. putida* KT2440 specifically, the **identified selenoprotein client is FdoG (PP_0489), a formate dehydrogenase major subunit bearing**

selenocysteine at residue 197, encoded in a *fdo* operon immediately upstream of *selAB* (see §3b).

- **Why the Sec matters (mechanistic payoff).** In Mo/W-containing formate dehydrogenases, "a selenocysteine (SeCys) or cysteine (Cys) ligand at the Mo atom in the active site is **essential for the reaction**" [PMID 25514355]. The *E. coli* FDH-H crystal structure shows the **molybdenum "directly coordinated to selenium"** and a mechanism "directly involv[ing] SeCys140 and His141 in proton abstraction" [PMID 9036855]. Thus SelA sits at the head of a precise mechanistic chain — **SelA → Sec-tRNA(Sec) → SelB-mediated UGA recoding → Sec-FdoG → Mo–Se catalytic center** — and its activity is a prerequisite for assembling a catalytically competent selenium-dependent formate dehydrogenase. (Note: the *fdo*-type enzyme is a cysteine/selenocysteine-ligand FDH; SelA is specifically required for the Sec-containing form.)

6. Evidence summary and confidence

- **Reaction/EC/cofactor:** Strong experimental evidence in orthologs (biochemical characterization of *E. coli* and archaeal/thermophilic enzymes) [PMID 1839607; 23266652; 23559248]. Confidence for *P. putida*: **High** (conserved family, HAMAP MF_00423).
- **Mechanism (PLP, dehydroalanine intermediate, selenophosphate donor):** Strong; supported by crystallography and enzymology of SelA and the parallel eukaryotic enzyme [PMID 23559248; 18093968; 17194211]. Confidence: **High**.
- **Quaternary structure / specificity determinants:** Direct structural + mutational evidence in *A. aeolicus* / *E. coli* SelA [PMID 23559248; 24456689; 23428847]. Confidence: **High** (family-conserved; *P. putida*-specific structure not solved).
- **Localization (cytoplasmic):** Inferred from sequence features, biochemistry, and pathway logic. Confidence: **High**.
- **Pathway/operon role:** Strong genetic and biochemical evidence [PMID 1839607; 12486013], plus confirmed *P. putida* genomic context (*selAB* = PP_0493–PP_0494 adjacent; *selD* = PP_0823; Sec-selenoprotein FdoG = PP_0489). Confidence: **High**.

7. Supported vs. refuted hypotheses

Supported 1. SslA is a PLP-dependent enzyme converting Ser-tRNA(Sec) → Sec-tRNA(Sec) (EC 2.9.1.1). ✓ 2. Selenophosphate is the selenium donor. ✓ 3. SslA is a ~500 kDa homodecamer (pentamer of dimers); oligomerization is essential for catalysis and tRNA(Sec) discrimination. ✓ 4. SslA is cytoplasmic and functions within the selA/selB/selC/selD pathway, feeding Sec-tRNA(Sec) to SelB for UGA recoding. ✓

Refuted / excluded 1. SslA acts on **free** serine or free selenocysteine — refuted; the substrate is the **tRNA-bound** seryl residue [PMID 1839607]. 2. SslA uses the eukaryotic-type **O-phosphoseryl** intermediate — refuted for bacteria; that phosphoserine route is specific to archaeal/eukaryotic **SepSecS** [PMID 18093968; 17194211]. 3. SslA is a small monomeric transferase — refuted; the functional unit is the decameric ring [PMID 24456689].

8. Limitations and future directions

- No biochemical or structural study of the *P. putida* KT2440 protein (PP_0493) itself exists; all mechanistic/structural detail is transferred from orthologs (*E. coli*, *A. aeolicus*) of a highly conserved family — justified here by 65% identity and a strictly conserved active site (§3c). A direct enzymatic assay and/or cryo-EM of PP_0493 would confirm strain-level parameters.
 - The *P. putida* selenoprotein complement identified from annotation is the FdoG-type formate dehydrogenase (PP_0489, Sec-197); the growth conditions under which it (and hence the SslA pathway) is required — e.g., anaerobic/microaerophilic formate metabolism and selenium availability — warrant targeted expression and knockout studies.
 - Whether *P. putida* SslA-dependent Sec incorporation is regulated by selenium availability via a SECIS-like leader (as in *E. coli*) is a testable, unresolved question.
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9. Key references (PMIDs)

- 1839607 — Forchhammer, Boesmler, Böck (1991): biochemical characterization of selenocysteine synthase (SslA) and SELB; selAB operon; PLP mechanism; substrate = Ser-tRNA(Sec).

- 23266652 — Manzine et al. (2013): *E. coli* SELA, EC 2.9.1.1, ~500 kDa PLP homodecamer converting seryl- to selenocysteyl-tRNA(Sec).
- 23559248 — Itoh et al. (2013, *Science*): decameric Sela-tRNA(Sec) ring structure; mechanism; D-arm-based substrate discrimination; PLP/Arg catalysis; independent evolution vs SepSecS.
- 24456689 — Itoh et al. (2014): fold-type-I PLP superfamily; pentamer-of-dimers assembly essential for active-site formation and specificity.
- 23428847 — Manzine et al. (2013): 1:1 Sela:tRNA(Sec) stoichiometry; D5 symmetry.
- 12486013 — Thanbichler & Böck (2002): SECIS/SelB UGA recoding; selAB operon regulation by selenium.
- 18093968 — Ganichkin et al. (2008): eukaryotic SecS structure/mechanism (contrast; phosphoserine route).
- 17194211 — Xu et al. (2007): eukaryotic Sec biosynthesis pathway; selenophosphate + phosphoseryl-tRNA substrates (contrast).
- 25514355 — Hartmann, Schwanhold & Leimkühler (2015): review of bacterial Mo/W formate dehydrogenases; Sec/Cys ligand at Mo essential for catalysis (downstream significance of Sela).
- 9036855 — Boyington et al. (1997, *Science*): FDH-H crystal structure; Sec140 directly coordinates Mo and participates in proton abstraction (downstream significance of Sela).